

# Integration Manual

for S32 DIO Driver

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## Chapter 1

### Revision History

| Revision | Date       | Author       | Description                                             |
|----------|------------|--------------|---------------------------------------------------------|
| 1.0      | 30.06.2023 | NXP RTD Team | S32 Real-Time Drivers AUTOSAR 4.4 Version 4.0.2 Release |

## Chapter 2

### Introduction

- [Supported Derivatives](#)
- [Overview](#)
- [About This Manual](#)
- [Acronyms and Definitions](#)
- [Reference List](#)

This integration manual describes the integration requirements for DIO Driver for S32 microcontrollers.

### 2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP Semiconductors:

- s32g274a\_bga525
- s32g254a\_bga525
- s32g233a\_bga525
- s32g234m\_bga525
- s32g378a\_bga525
- s32g379a\_bga525
- s32g398a\_bga525
- s32g399a\_bga525
- s32g338m\_bga525
- s32g339m\_bga525
- s32g358a\_bga525
- s32g359a\_bga525
- s32r45\_bga780

All of the above microcontroller devices are collectively named as S32.

## 2.2 Overview

**AUTOSAR (AUTomotive Open System ARchitecture)** is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR:

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

## 2.3 About This Manual

This Technical Reference employs the following typographical conventions:

- **Boldface** style: Used for important terms, notes and warnings.
- *Italic* style: Used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

Warning

This is a warning

## 2.4 Acronyms and Definitions

| Term  | Definition                               |
|-------|------------------------------------------|
| API   | Application Programming Interface        |
| ASM   | Assembler                                |
| BSMI  | Basic Software Make file Interface       |
| CAN   | Controller Area Network                  |
| C/CPP | C and C++ Source Code                    |
| CS    | Chip Select                              |
| CTU   | Cross Trigger Unit                       |
| DEM   | Diagnostic Event Manager                 |
| DET   | Development Error Tracer                 |
| DMA   | Direct Memory Access                     |
| ECU   | Electronic Control Unit                  |
| FIFO  | First In First Out                       |
| LSB   | Least Significant Bit                    |
| MCU   | Micro Controller Unit                    |
| MIDE  | Multi Integrated Development Environment |
| MSB   | Most Significant Bit                     |
| N/A   | Not Applicable                           |
| RAM   | Random Access Memory                     |
| SIU   | Systems Integration Unit                 |
| SWS   | Software Specification                   |
| VLE   | Variable Length Encoding                 |
| XML   | Extensible Markup Language               |

## 2.5 Reference List

| # | Title                       | Version                                         |
|---|-----------------------------|-------------------------------------------------|
| 1 | Specification of Dio Driver | AUTOSAR Release 4.4.0                           |
| 2 | Reference Manual            | S32G2 Reference Manual, Rev. 7, February 2023   |
|   |                             | S32G3 Reference Manual, Rev.3, Feb 2023         |
|   |                             | S32R45 Reference Manual, Rev. 3, 12/2021        |
| 3 | Datasheet                   | S32G2 Data Sheet, Rev 6, Dec 2022               |
|   |                             | S32G3 Data Sheet, Rev 2, RC, Feb 2023           |
|   |                             | VR5510 Data Sheet, Rev 5, April 2022            |
|   |                             | S32R45 Data Sheet, Rev. 3, RC — 11/2022         |
| 4 | Errata                      | S32G2: Mask Set Errata for Mask 0P77B v3.0      |
|   |                             | S32G2: Mask Set Errata for Mask 1P77B v1.0      |
|   |                             | S32G3: Mask Set Errata for Mask 1P72B v2.1      |
|   |                             | S32R45: Mask Set Errata for Mask P57D, Rev. 2.0 |

## Chapter 3

### Building the driver

- [Build Options](#)
- [Files required for compilation](#)
- [Setting up the plugins](#)

This section describes the source files and various compilers, linker options used for building the driver. It also explains the EB Tresos Studio plugin setup procedure.

#### 3.1 Build Options

- [GCC Compiler/Assembler/Linker Options](#)
- [GHS Compiler/Assembler/Linker Options](#)
- [DIAB Compiler/Assembler/Linker Options](#)

The RTD driver files are compiled using:

- NXP GCC 9.2.0 20190812 (Build 1649 Revision gaf57174)
- Green Hills Multi 7.1.6d / Compiler 2020.1.4
- Wind River Diab Compiler 7.0.3

The compiler, assembler, and linker flags used for building the driver are explained below.

The TS\_T40D11M40I2R0 part of the plugin name is composed as follows:

- T = Target\_Id (e.g. T40 identifies Cortex-M architecture)
- D = Derivative\_Id (e.g. D11 identifies S32 platform)
- M = SW\_Version\_Major and SW\_Version\_Minor
- I = SW\_Version\_Patch
- R = Reserved

##### 3.1.1 GCC Compiler/Assembler/Linker Options

###### 3.1.1.1 GCC Compiler Options



| Compiler Option                       | Description                                                                                                                                                                                                                        |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -mcpu=cortex-m7                       | Targeted ARM processor for which GCC should tune the performance of the code                                                                                                                                                       |
| -mthumb                               | Generates code that executes in Thumb state                                                                                                                                                                                        |
| -mlittle-endian                       | Generate code for a processor running in little-endian mode                                                                                                                                                                        |
| -mfpv=fpv5-sp-d16                     | Specifies the floating-point hardware available on the target                                                                                                                                                                      |
| -mfloat-abi=hard                      | Specifies the floating-point ABI to use. "hard" allows generation of floating-point instructions and uses FPU-specific calling conventions                                                                                         |
| -std=c99                              | Specifies the ISO C99 base standard                                                                                                                                                                                                |
| -Os                                   | Optimize for size. Enables all -O2 optimizations except those that often increase code size                                                                                                                                        |
| -ggdb3                                | Produce debugging information for use by GDB using the most expressive format available, including GDB extensions if at all possible. Level 3 includes extra information, such as all the macro definitions present in the program |
| -Wall                                 | Enables all the warnings about constructions that some users consider questionable, and that are easy to avoid (or modify to prevent the warning), even in conjunction with macros                                                 |
| -Wextra                               | This enables some extra warning flags that are not enabled by -Wall                                                                                                                                                                |
| -pedantic                             | Issue all the warnings demanded by strict ISO C. Reject all programs that use forbidden extensions. Follows the version of the ISO C standard specified by the aforementioned -std option                                          |
| -Wstrict-prototypes                   | Warn if a function is declared or defined without specifying the argument types                                                                                                                                                    |
| -Wundef                               | Warn if an undefined identifier is evaluated in an #if directive. Such identifiers are replaced with zero                                                                                                                          |
| -Wunused                              | Warn whenever a function, variable, label, value, macro is unused                                                                                                                                                                  |
| -Werror=implicit-function-declaration | Make the specified warning into an error. This option throws an error when a function is used before being declared                                                                                                                |
| -Wsign-compare                        | Warn when a comparison between signed and unsigned values could produce an incorrect result when the signed value is converted to unsigned.                                                                                        |
| -Wdouble-promotion                    | Give a warning when a value of type float is implicitly promoted to double                                                                                                                                                         |
| -fno-short-enums                      | Specifies that the size of an enumeration type is at least 32 bits regardless of the size of the enumerator values.                                                                                                                |
| -funsigned-char                       | Let the type char be unsigned by default, when the declaration does not use either signed or unsigned                                                                                                                              |
| -funsigned-bitfields                  | Let a bit-field be unsigned by default, when the declaration does not use either signed or unsigned                                                                                                                                |
| -fomit-frame-pointer                  | Omit the frame pointer in functions that don't need one. This avoids the instructions to save, set up and restore the frame pointer; on many targets it also makes an extra register available.                                    |

| Compiler Option                 | Description                                                                                                                                                                                                                                                                           |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -fno-common                     | Makes the compiler place uninitialized global variables in the BSS section of the object file. This inhibits the merging of tentative definitions by the linker so you get a multiple-definition error if the same variable is accidentally defined in more than one compilation unit |
| -fstack-usage                   | This option is only used to build test for generation Ram/↔ Stack size report. Makes the compiler output stack usage information for the program, on a per-function basis                                                                                                             |
| -fdump-ipa-all                  | This option is only used to build test for generation Ram/↔ Stack size report. Enables all inter-procedural analysis dumps                                                                                                                                                            |
| -c                              | Stop after assembly and produce an object file for each source file                                                                                                                                                                                                                   |
| -DS32XX                         | Predefine S32XX as a macro, with definition 1                                                                                                                                                                                                                                         |
| -DS32G2XX                       | Predefine S32G2XX as a macro, with definition 1                                                                                                                                                                                                                                       |
| -DS32G3XX                       | Predefine S32G3XX as a macro, with definition 1                                                                                                                                                                                                                                       |
| -DS32R45                        | Predefine S32R45 as a macro, with definition 1                                                                                                                                                                                                                                        |
| -DGCC                           | Predefine GCC as a macro, with definition 1                                                                                                                                                                                                                                           |
| -DUSE_SW_VECTOR_MODE            | Predefine USE_SW_VECTOR_MODE as a macro, with definition 1. By default, the drivers are compiled to handle interrupts in Software Vector Mode                                                                                                                                         |
| -DD_CACHE_ENABLE                | Predefine D_CACHE_ENABLE as a macro, with definition 1. Enables data cache initialization in source file system.↔ c under the Platform driver                                                                                                                                         |
| -DI_CACHE_ENABLE                | Predefine I_CACHE_ENABLE as a macro, with definition 1. Enables instruction cache initialization in source file system.c under the Platform driver                                                                                                                                    |
| -DENABLE_FPU                    | Predefine ENABLE_FPU as a macro, with definition 1. Enables FPU initialization in source file system.c under the Platform driver                                                                                                                                                      |
| -DMCAL_ENABLE_USER_MODE_SUPPORT | Predefine MCAL_ENABLE_USER_MODE_SUPPORT↔ RT as a macro, with definition 1. Allows drivers to be configured in user mode.                                                                                                                                                              |

### 3.1.1.2 GCC Assembler Options

| Assembler Option      | Description                                                                                                                                |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| -X assembler-with-cpp | Specifies the language for the following input files (rather than letting the compiler choose a default based on the file name suffix)     |
| -mcpu=cortexm7        | Targeted ARM processor for which GCC should tune the performance of the code                                                               |
| -mfpu=fpv5-sp-d16     | Specifies the floating-point hardware available on the target                                                                              |
| -mfloat-abi=hard      | Specifies the floating-point ABI to use. "hard" allows generation of floating-point instructions and uses FPU-specific calling conventions |
| -mthumb               | Generates code that executes in Thumb state                                                                                                |
| -c                    | Stop after assembly and produce an object file for each source file                                                                        |

### 3.1.1.3 GCC Linker Options

| Linker Option        | Description                                                                                                                                                                                                                        |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -Wl,-Map,filename    | Produces a map file                                                                                                                                                                                                                |
| -T linkerfile        | Use linkerfile as the linker script. This script replaces the default linker script (rather than adding to it)                                                                                                                     |
| -entry=Reset_Handler | Specifies that the program entry point is Reset_Handler                                                                                                                                                                            |
| -nostartfiles        | Do not use the standard system startup files when linking                                                                                                                                                                          |
| -mcpu=cortexm7       | Targeted ARM processor for which GCC should tune the performance of the code                                                                                                                                                       |
| -mthumb              | Generates code that executes in Thumb state                                                                                                                                                                                        |
| -mfpu=fpv5-sp-d16    | Specifies the floating-point hardware available on the target                                                                                                                                                                      |
| -mfloat-abi=hard     | Specifies the floating-point ABI to use. "hard" allows generation of floating-point instructions and uses FPU-specific calling conventions                                                                                         |
| -mlittle-endian      | Generate code for a processor running in little-endian mode                                                                                                                                                                        |
| -ggdb3               | Produce debugging information for use by GDB using the most expressive format available, including GDB extensions if at all possible. Level 3 includes extra information, such as all the macro definitions present in the program |
| -lc                  | Link with the C library                                                                                                                                                                                                            |
| -lm                  | Link with the Math library                                                                                                                                                                                                         |
| -lgcc                | Link with the GCC library                                                                                                                                                                                                          |

### 3.1.2 GHS Compiler/Assembler/Linker Options

#### 3.1.2.1 GHS Compiler Options

| Compiler Option | Description                                                                                                                                                                                                  |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -cpu=cortexm7   | Selects target processor: Arm Cortex M7                                                                                                                                                                      |
| -thumb          | Selects generating code that executes in Thumb state                                                                                                                                                         |
| -fpu=vfpv5_d16  | Specifies hardware floating-point using the v5 version of the VFP instruction set, with 16 double-precision floating-point registers                                                                         |
| -fsingle        | Use hardware single-precision, software double-precision FP instructions                                                                                                                                     |
| -C99            | Use (strict ISO) C99 standard (without extensions)                                                                                                                                                           |
| -ghstd=last     | Use the most recent version of Green Hills Standard mode (which enables warnings and errors that enforce a stricter coding standard than regular C and C++)                                                  |
| -Osize          | Optimize for size                                                                                                                                                                                            |
| -gnu_asm        | Enables GNU extended asm syntax support                                                                                                                                                                      |
| -dual_debug     | Generate DWARF 2.0 debug information                                                                                                                                                                         |
| -G              | Generate debug information                                                                                                                                                                                   |
| -keeptempfiles  | Prevents the deletion of temporary files after they are used. If an assembly language file is created by the compiler, this option will place it in the current directory instead of the temporary directory |

| Compiler Option                 | Description                                                                                                                                        |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| -Wimplicit-int                  | Produce warnings if functions are assumed to return int                                                                                            |
| -Wshadow                        | Produce warnings if variables are shadowed                                                                                                         |
| -Wtrigraphs                     | Produce warnings if trigraphs are detected                                                                                                         |
| -Wundef                         | Produce a warning if undefined identifiers are used in #if preprocessor statements                                                                 |
| -unsigned_chars                 | Let the type char be unsigned, like unsigned char                                                                                                  |
| -unsigned_fields                | Bitfields declared with an integer type are unsigned                                                                                               |
| -no_commons                     | Allocates uninitialized global variables to a section and initializes them to zero at program startup                                              |
| -no_exceptions                  | Disables C++ support for exception handling                                                                                                        |
| -no_slash_comment               | C++ style // comments are not accepted and generate errors                                                                                         |
| -prototype_errors               | Controls the treatment of functions referenced or called when no prototype has been provided                                                       |
| -incorrect_pragma_warnings      | Controls the treatment of valid #pragma directives that use the wrong syntax                                                                       |
| -c                              | Stop after assembly and produce an object file for each source file                                                                                |
| -DS32XX                         | Predefine S32XX as a macro, with definition 1                                                                                                      |
| -DS32G2XX                       | Predefine S32G2XX as a macro, with definition 1                                                                                                    |
| -DS32G3XX                       | Predefine S32G3XX as a macro, with definition 1                                                                                                    |
| -DS32R45                        | Predefine S32R45 as a macro, with definition 1                                                                                                     |
| -DGHS                           | Predefine GHS as a macro, with definition 1                                                                                                        |
| -DUSE_SW_VECTOR_MODE            | Predefine USE_SW_VECTOR_MODE as a macro, with definition 1. By default, the drivers are compiled to handle interrupts in Software Vector Mode      |
| -DD_CACHE_ENABLE                | Predefine D_CACHE_ENABLE as a macro, with definition 1. Enables data cache initialization in source file system.↵c under the Platform driver       |
| -DI_CACHE_ENABLE                | Predefine I_CACHE_ENABLE as a macro, with definition 1. Enables instruction cache initialization in source file system.c under the Platform driver |
| -DENABLE_FPU                    | Predefine ENABLE_FPU as a macro, with definition 1. Enables FPU initialization in source file system.c under the Platform driver                   |
| -DMCAL_ENABLE_USER_MODE_SUPPORT | Predefine MCAL_ENABLE_USER_MODE_SUPPORT↵RT as a macro, with definition 1. Allows drivers to be configured in user mode                             |

### 3.1.2.2 GHS Assembler Options

| Assembler Option           | Description                                                                                                                          |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| -cpu=cortexm7              | Selects target processor: Arm Cortex M7                                                                                              |
| -fpu=vfpv5_d16             | Specifies hardware floating-point using the v5 version of the VFP instruction set, with 16 double-precision floating-point registers |
| -fsingle                   | Use hardware single-precision, software double-precision FP instructions                                                             |
| -preprocess_assembly_files | Controls whether assembly files with standard extensions such as .s and .asm are preprocessed                                        |

| Assembler Option | Description                                                                                  |
|------------------|----------------------------------------------------------------------------------------------|
| -list            | Creates a listing by using the name and directory of the object file with the .lst extension |
| -c               | Stop after assembly and produce an object file for each source file                          |

### 3.1.2.3 GHS Linker Options

| Linker Option            | Description                                                                                                                                                                                                        |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -e Reset_Handler         | Make the symbol Reset_Handler be treated as a root symbol and the start label of the application                                                                                                                   |
| -T linker_script_file.ld | Use linker_script_file.ld as the linker script. This script replaces the default linker script (rather than adding to it)                                                                                          |
| -map                     | Produce a map file                                                                                                                                                                                                 |
| -keepmap                 | Controls the retention of the map file in the event of a link error                                                                                                                                                |
| -Mn                      | Generates a listing of symbols sorted alphabetically/numerically by address                                                                                                                                        |
| -delete                  | Instructs the linker to remove functions that are not referenced in the final executable. The linker iterates to find functions that do not have relocations pointing to them and eliminates them                  |
| -ignore_debug_references | Ignores relocations from DWARF debug sections when using -delete. DWARF debug information will contain references to deleted functions that may break some third-party debuggers                                   |
| -Llibrary_path           | Points to library_path (the libraries location) for thumb2 to be used for linking                                                                                                                                  |
| -larch                   | Link architecture specific library                                                                                                                                                                                 |
| -lstartup                | Link run-time environment startup routines. The source code for the modules in this library is provided in the src/libstartup directory                                                                            |
| -lind_sd                 | Link language-independent library, containing support routines for features such as software floating point, run-time error checking, C99 complex numbers, and some general purpose routines of the ANSI C library |
| -v                       | Prints verbose information about the activities of the linker, including the libraries it searches to resolve undefined symbols                                                                                    |
| -nostartfiles            | Controls the start files to be linked into the executable                                                                                                                                                          |

### 3.1.3 DIAB Compiler/Assembler/Linker Options

#### 3.1.3.1 DIAB Compiler Options

| Compiler Option        | Description                                                                                    |
|------------------------|------------------------------------------------------------------------------------------------|
| -tARMCORTEXM7MG:simple | Selects target processor (hardware single-precision, software double-precision floating-point) |
| -mthumb                | Selects generating code that executes in Thumb state                                           |
| -std=c99               | Follows the C99 standard for C                                                                 |
| -Oz                    | Like -O2 with further optimizations to reduce code size                                        |
| -g                     | Generates DWARF 4.0 debug information                                                          |
| -fstandalone-debug     | Emits full debug info for all types used by the program                                        |

| Compiler Option                       | Description                                                                                                                                        |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| -Wstrict-prototypes                   | Warn if a function is declared or defined without specifying the argument types                                                                    |
| -Wsign-compare                        | Produce warnings when comparing signed type with unsigned type                                                                                     |
| -Wdouble-promotion                    | Give a warning when a value of type float is implicitly promoted to double                                                                         |
| -Wunknown-pragmas                     | Issues a warning for unknown pragmas                                                                                                               |
| -Wundef                               | Warns if an undefined identifier is evaluated in an #if directive. Such identifiers are replaced with zero                                         |
| -Wextra                               | Enables some extra warning flags that are not enabled by '-Wall'                                                                                   |
| -Wall                                 | Enables all of the most useful warnings (for historical reasons this option does not literally enable all warnings)                                |
| -pedantic                             | Emits a warning whenever the standard specified by the -std option requires a diagnostic                                                           |
| -Werror=implicit-function-declaration | Generates an error whenever a function is used before being declared                                                                               |
| -fno-common                           | Compile common globals like normal definitions                                                                                                     |
| -fno-signed-char                      | Char is unsigned                                                                                                                                   |
| -fno-trigraphs                        | Do not process trigraph sequences                                                                                                                  |
| -V                                    | Displays the current version number of the tool suite                                                                                              |
| -c                                    | Stop after assembly and produce an object file for each source file                                                                                |
| -DS32XX                               | Predefine S32XX as a macro, with definition 1                                                                                                      |
| -DS32G2XX                             | Predefine S32G2XX as a macro, with definition 1                                                                                                    |
| -DS32G3XX                             | Predefine S32G3XX as a macro, with definition 1                                                                                                    |
| -DS32R45                              | Predefine S32R45 as a macro, with definition 1                                                                                                     |
| -DDIAB                                | Predefine DIAB as a macro, with definition 1                                                                                                       |
| -DUSE_SW_VECTOR_MODE                  | Predefine USE_SW_VECTOR_MODE as a macro, with definition 1. By default, the drivers are compiled to handle interrupts in Software Vector Mode      |
| -DD_CACHE_ENABLE                      | Predefine D_CACHE_ENABLE as a macro, with definition 1. Enables data cache initialization in source file system.c under the Platform driver        |
| -DI_CACHE_ENABLE                      | Predefine I_CACHE_ENABLE as a macro, with definition 1. Enables instruction cache initialization in source file system.c under the Platform driver |
| -DENABLE_FPU                          | Predefine ENABLE_FPU as a macro, with definition 1. Enables FPU initialization in source file system.c under the Platform driver                   |
| -DMCAL_ENABLE_USER_MODE_SUPPORT       | Predefine MCAL_ENABLE_USER_MODE_SUPPORT as a macro, with definition 1. Allows drivers to be configured in user mode                                |

### 3.1.3.2 DIAB Assembler Options

| Assembler Option      | Description                                                           |
|-----------------------|-----------------------------------------------------------------------|
| -mthumb               | Selects generating code that executes in Thumb state                  |
| -Xpreprocess-assembly | Invokes C preprocessor on assembly files before running the assembler |
| -Xassembly-listing    | Produces an .lst assembly listing file                                |
| -c                    | Stop after assembly and produce an object file for each source file   |

### 3.1.3.3 DIAB Linker Options

| Linker Option          | Description                                                                                                                |
|------------------------|----------------------------------------------------------------------------------------------------------------------------|
| -e Reset_Handler       | Make the symbol Reset_Handler be treated as a root symbol and the start label of the application                           |
| linker_script_file.dld | Use linker_script_file.dld as the linker script. This script replaces the default linker script (rather than adding to it) |
| -m30                   | m2 + m4 + m8 + m16                                                                                                         |
| -Xstack-usage          | Gathers and display stack usage at link time                                                                               |
| -Xpreprocess-lecl      | Perform pre-processing on linker scripts                                                                                   |
| -Llibrary_path         | Points to the libraries location for ARMV7EMMG to be used for linking                                                      |
| -lc                    | Links with the standard C library                                                                                          |
| -lm                    | Links with the math library                                                                                                |

## 3.2 Files required for compilation

This section describes the include files required to compile, assemble (if assembler code) and link the DIO driver for S32 microcontrollers.

To avoid integration of incompatible files, all the include files from other modules shall have the same AR\_MAJOR\_VERSION and AR\_MINOR\_VERSION, i.e. only files with the same AUTOSAR major and minor versions can be compiled.

### Dio Files

- ..\Dio\_TS\_T40D11M40I2R0\include\Dio.h
- ..\Dio\_TS\_T40D11M40I2R0\include\Dio\_Ipw.h
- ..\Dio\_TS\_T40D11M40I2R0\include\Siul2\_Dio\_Ip.h
- ..\Dio\_TS\_T40D11M40I2R0\src\Dio.c
- ..\Dio\_TS\_T40D11M40I2R0\src\Dio\_Ipw.c
- ..\Dio\_TS\_T40D11M40I2R0\src\Siul2\_Dio\_Ip.c

### Dio Generated Files

- Dio\_Cfg.c - This file should be generated by the user using a configuration tool for compilation.

## Building the driver

- Dio\_Cfg.h - This file should be generated by the user using a configuration tool for compilation.
- Siul2\_Dio\_Ip\_Cfg.h - This file should be generated by the user using a configuration tool for compilation.

### As a deviation from standard:

- Dio\_Cfg.c - This file will contain the definition for all configuration structures containing only variables that are not variant aware, configured and generated only once. This file alone does not contain the whole structure needed by Port\_Init function to configure the driver. Based on the number of variants configured in the EcuC, there can be more than one configuration structure for one module even for PreCompile variant.

### Files from BaseNXP common folder

- ..\BaseNXP\_TS\_T40D11M40I2R0\include\
- ..\BaseNXP\_TS\_T40D11M40I2R0\header\
- ..\BaseNXP\_TS\_T40D11M40I2R0\src\OsIf\_Timer.c
- ..\BaseNXP\_TS\_T40D11M40I2R0\src\OsIf\_Timer\_System.c

### Files from Det folder:

- ..\Det\_TS\_T40D11M40I2R0\include\Det.h
- ..\Det\_TS\_T40D11M40I2R0\src\Det.c

### Files from Rte folder:

- ..\Rte\_TS\_T40D11M40I2R0\include\SchM\_Dio.h
- ..\Rte\_TS\_T40D11M40I2R0\src\SchM\_Dio.c

### Files from Os folder:

- ..\Os\_TS\_T40D11M40I2R0\include\Os.h



### 3.3 Setting up the plugins

The DIO driver was designed to be configured by using the EB Tresos Studio (version EB tresos Studio 27.1.0 or later.)

**Location of various files inside the DIO module folder:**

- VSMD (Vendor Specific Module Definition) file in EB tresos Studio XDM format:
  - ..\Dio\_TS\_T40D11M40I2R0\config\Dio.xdm
- VSMD (Vendor Specific Module Definition) file(s) in AUTOSAR compliant EPD format:
  - ..\Dio\_TS\_T40D11M40I2R0\autosar\Dio\_s32g233a\_bga525.epd
  - ..\Dio\_TS\_T40D11M40I2R0\autosar\Dio\_s32g234m\_bga525.epd
  - ..\Dio\_TS\_T40D11M40I2R0\autosar\Dio\_s32g254a\_bga525.epd
  - ..\Dio\_TS\_T40D11M40I2R0\autosar\Dio\_s32g274a\_bga525.epd
  - ..\Dio\_TS\_T40D11M40I2R0\autosar\Dio\_s32g378a\_bga525.epd
  - ..\Dio\_TS\_T40D11M40I2R0\autosar\Dio\_s32g379a\_bga525.epd
  - ..\Dio\_TS\_T40D11M40I2R0\autosar\Dio\_s32g398a\_bga525.epd
  - ..\Dio\_TS\_T40D11M40I2R0\autosar\Dio\_s32g399a\_bga525.epd
  - ..\Dio\_TS\_T40D11M40I2R0\autosar\Dio\_s32g338m\_bga525.epd
  - ..\Dio\_TS\_T40D11M40I2R0\autosar\Dio\_s32g339m\_bga525.epd
  - ..\Dio\_TS\_T40D11M40I2R0\autosar\Dio\_s32g358a\_bga525.epd
  - ..\Dio\_TS\_T40D11M40I2R0\autosar\Dio\_s32g359a\_bga525.epd
  - ..\Dio\_TS\_T40D11M40I2R0\autosar\Dio\_s32r45\_bga780.epd
- Code Generation Templates for parameters without variation points:
  - ..\Dio\_TS\_T40D11M40I2R0\generate\_PC\include\Dio\_Cfg.h
  - ..\Dio\_TS\_T40D11M40I2R0\generate\_PC\include\Siul2\_Dio\_Ip\_Cfg.h
  - ..\Dio\_TS\_T40D11M40I2R0\generate\_PC\src\Dio\_Cfg.c

**Steps to generate the configuration:**

1. Copy the module folders Dio\_TS\_T40D11M40I2R0, BaseNXP\_TS\_T40D11M40I2R0, Resource\_TS\_T40D11M40I2R0, Det\_TS\_T40D11M40I2R0, Rte\_TS\_T40D11M40I2R0, Ecuc\_TS\_T40D11M40I2R0, Os\_TS\_T40D11M40I2R0 into the Tresos plugins folder.
2. Set the desired Tresos Output location folder for the generated sources and header files.
3. Use the EB tresos Studio GUI to modify ECU configuration parameters values.
4. Generate the configuration files.



## Chapter 4

### Function calls to module

- [Function Calls during Start-up](#)
- [Function Calls during Shutdown](#)
- [Function Calls during Wake-up](#)

#### 4.1 Function Calls during Start-up

None.

#### 4.2 Function Calls during Shutdown

None.

#### 4.3 Function Calls during Wake-up

None.

## Chapter 5

### Module requirements

- Exclusive areas to be defined in BSW scheduler
- Exclusive areas not available on this platform
- Peripheral Hardware Requirements
- ISR to configure within AutosarOS - dependencies
- ISR Macro
- Other AUTOSAR modules - dependencies
- Data Cache Restrictions
- User Mode support
- Multicore support

#### 5.1 Exclusive areas to be defined in BSW scheduler

In the current implementation, DIO is using the services of Run-Time Environment (RTE) for entering and exiting the critical regions. RTE implementation is done by the integrators of the RTD using OS or non-OS services. For testing the Dio driver, stubs are used for RTE. The following critical regions are used in the DIO driver:

**DIO\_EXCLUSIVE\_AREA\_00** is used in function Dio\_FlipChannel. It protect the SIUL2\_GPDO\_A←DDR32 and SIUL2\_GPDI\_ADDR32 register from read/modify/write operation.

**DIO\_EXCLUSIVE\_AREA\_01** is used in function Siul2\_Dio\_Ip\_WritePin. It protect the GPDO register from read/modify/write operation.

##### Critical Region Exclusive Matrix

Below is the table depicting the exclusivity between different critical region IDs from the DIO driver. If there is an “X” in the table, it means that those 2 critical regions cannot interrupt each other.

| #         | DIO_EA_00 | DIO_EA_01 |
|-----------|-----------|-----------|
| DIO_EA_00 | x         |           |
| DIO_EA_01 |           | x         |

### Note

DIO\_EA\_xx means DIO\_EXCLUSIVE\_AREA\_xx

## 5.2 Exclusive areas not available on this platform

None

## 5.3 Peripheral Hardware Requirements

The Dio driver uses SIUL2 peripheral.

Port pins that are available on a particular package are described in the S32 Reference Manual.

The formula for calculating port and channel number for PINx is:

$PORT = PINx / 16$

$CHANNEL = PINx \% 16$

## 5.4 ISR to configure within AutosarOS - dependencies

Not implemented.

## 5.5 ISR Macro

RTD drivers use the ISR macro to define the functions that will process hardware interrupts. Depending on whether the OS is used or not, this macro can have different definitions.

**5.5.1 Without an Operating System** The macro *USING\_OS\_AUTOSAROS* must not be defined.

### 5.5.1.1 Using Software Vector Mode

The macro *USE\_SW\_VECTOR\_MODE* must be defined and the ISR macro is defined as:

```
#define ISR(IsrName) void IsrName(void)
```

In this case, the drivers' interrupt handlers are normal C functions and their prologue/epilogue will handle the context save and restore.

### 5.5.1.2 Using Hardware Vector Mode

The macro *USE\_SW\_VECTOR\_MODE* must not be defined and the ISR macro is defined as:

```
#define ISR(IsrName) INTERRUPT_FUNC void IsrName(void)
```

In this case, the drivers' interrupt handlers must also handle the context save and restore.

**5.5.2 With an Operating System** Please refer to your OS documentation for description of the ISR macro.

## 5.6 Other AUTOSAR modules - dependencies

- **PORT:** The PORT module is used to configure the port pins with the needed modes, before they are used by the DIO - module.
- **DET:** The DET module is used for enabling Development error detection. The API function used is `Det_ReportError()`. The activation / deactivation of Development error detection is configurable using the `DioDevErrorDetect` configuration parameter.
- **BASENXP:** The BASENXP module contains the common files/definitions needed by all RTD modules.
- **RESOURCE:** The RESOURCE module is used to select microcontroller derivatives.
- **RTE:** The RTE module is used to manage the exclusive area inside DIO driver.
- **ECUC:** The ECUC module is used for ECU configuration. RTD modules need ECUC to retrieve the variant information.
- **Os:** The OS module is used for OS configuration. RTD modules need OS to retrieve the application information.
- **MCU:** The MCU driver provides services for basic microcontroller initialization, power down functionality, reset and microcontroller specific functions required by other RTD software modules. The clocks need to be initialized prior to using the DIO driver

## 5.7 Data Cache Restrictions

None.

## 5.8 User Mode support

- [User Mode configuration in the module](#)
- [User Mode configuration in AutosarOS](#)

### 5.8.1 User Mode configuration in the module

The Dio driver can be run from user mode. Dio driver depends on Port when it comes to configuration of its capability to run in user mode. The user mode (Port Enable User Mode Support) should be enabled in Port plugin in order for Dio to be able to run its code in user mode.

Please note that according with Dio external assumption SWS\_Dio\_00102, "The Dio module's user shall only use the Dio functions after the Port driver has been initialized. Otherwise the Dio module will exhibit undefined behavior."

### 5.8.2 User Mode configuration in AutosarOS

When User mode is enabled, the driver may have the functions that need to be called as trusted functions in AutosarOS context. Those functions are already defined in driver and declared in the header `<IpName>_Ip_TrustedFunctions.h`. This header also included all headers files that contains all types definition used by parameters or return types of those functions. Refer the chapter [User Mode configuration in the module](#) for more detail about those functions and the name of header files they are declared inside. Those functions will be called indirectly with the naming convention below in order to AutosarOS can call them as trusted functions.

```
Call_<Function_Name>_TRUSTED(parameter1,parameter2,...)
```

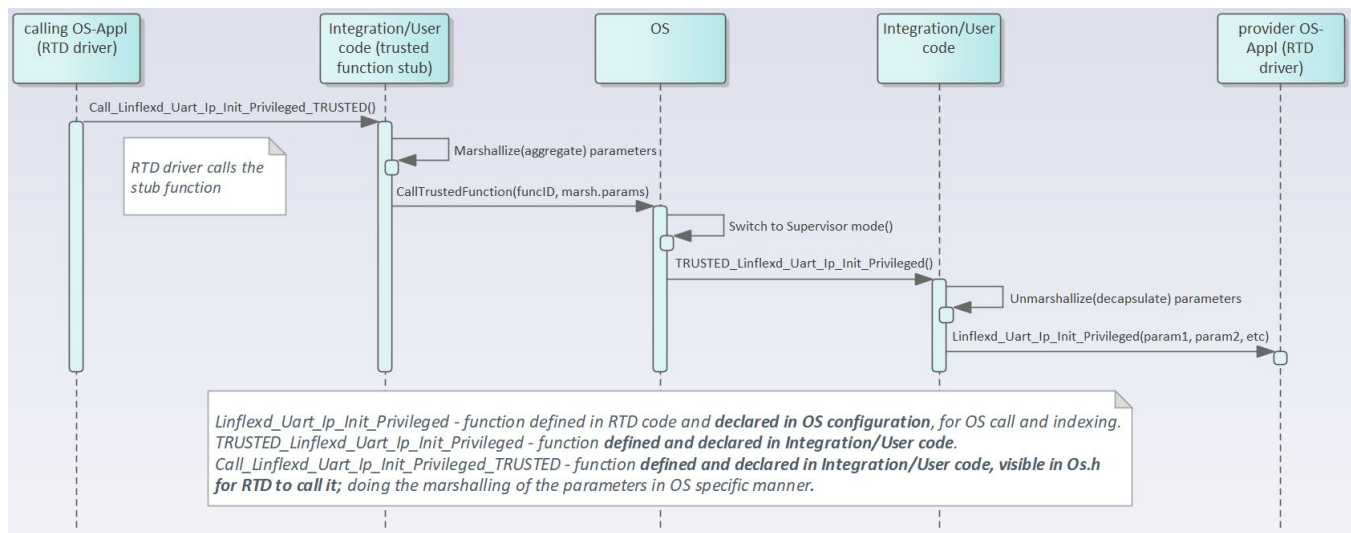
That is the result of macro expansion `OsIf_Trusted_Call` in driver code:

```
#define OsIf_Trusted_Call[1-6params](name,param1,...,param6) Call_##name##_TRUSTED(param1,...,param6)
```

So, the following steps need to be done in AutosarOS:

- Ensure `MCAL_ENABLE_USER_MODE_SUPPORT` macro is defined in the build system or somewhere global.
- Define and declare all functions that need to call as trusted functions follow the naming convention above in Integration/User code. They need to be visible in `Os.h` for the driver to call them. They will do the marshalling of the parameters and call `CallTrustedFunction()` in OS specific manner.
- `CallTrustedFunction()` will switch to privileged mode and call `TRUSTED_<Function_Name>()`.
- `TRUSTED_<Function_Name>()` function is also defined and declared in Integration/User code. It will un-marshalling of the parameters to call `<Function_Name>()` of driver. The `<Function_Name>()` functions are already defined in driver and declared in `<IpName>_Ip_TrustedFunctions.h`. This header should be included in OS for OS call and indexing these functions.

See the sequence chart below for an example calling `Linflexd_Uart_Ip_Init_Privileged()` as a trusted function.



**Figure 5.1 Example sequence chart for calling `Linflexd_Uart_Ip_Init_Privileged` as trusted function**

## 5.9 Multicore support

The **Dio** implements the "**Autosar 4.4 MCAL Multicore Distribution**" according to type III, in which the mappable element is set to Channels. For additional details, please refer to **AUTOSAR\_EXP\_BSWDistribution**↵  
**Guide**.

The **Dio** and the mappable elements can be allocated to zero, one or several ECUC partitions, by means of "**↵DioEcucPartionRef**". If the Dio is mapped to zero ECUC partitions, the **Dio** behavior reverts to single-core implementation, similar to previous Autosar versions. If the **Dio** is mapped to one or more ECUC partitions, the Dio enforces the following multi-core assumptions:

1. The Dio assumes there is a single EcucPartition allocated per core. Internally, the module will use the Core ID returned by GetCoreID API to reference the appropriate global data and configuration elements.
2. The Dio assumes the EcucCoreIDs are defined in a compact/consecutive order, starting from zero. The rationale is that the number of EcucPartitions is used for dimensioning the Dio internal variables and the EcucCoreIDs are used for indexing those variables. (AR-86601 Zero based and dense IDs for OS-Cores and OS-Applications)
3. The Dio assumes that initialization is performed on a single, designated core. It is called only once, with a single configuration structure(Type III).
4. The Dio initialization expects the upper layer will pass the correct initialization pointer, specific to the partition in which the driver is to be used. For example: EcucPartition\_1 is assigned to CoreID 1; It will be called with configuration structure init, on Core 1.
5. The Dio will check upon each API call if the requested resource is configured to be available on the current core, if DET error reporting is enabled.
6. The Dio requires that all variables in NonCacheable MemMap sections be allocated accordingly, to avoid data corruption in multicore context.
7. The Dio assumes that RTE module implements the EXCLUSIVE AREAS to be coreaware only. The rationale is that the module implementation ensures data integrity by separating the mappable elements for different cores already, thus implementing the EXCLUSIVE AREAS in a blocking manner (ex: spin-lock) on a multicore scope, might affect the performance of the drivers on the two cores, although they might access separate HW elements. For single-core scope, the EXCLUSIVE AREAS keep the same purpose as on previous AUTOSAR implementations. (to be updated per Dio usecase, to be detailed/removed if some modules require such kind of functionality for critical features which cannot be atomically shared among cores)



## Chapter 6

### Main API Requirements

- [Main function calls within BSW scheduler](#)
- [API Requirements](#)
- [Calls to Notification Functions, Callbacks, Callouts](#)

#### 6.1 Main function calls within BSW scheduler

None.

#### 6.2 API Requirements

None.

#### 6.3 Calls to Notification Functions, Callbacks, Callouts

None.



## Chapter 7

### Memory allocation

- [Sections to be defined in Dio\\_MemMap.h](#)
- [Linker command file](#)

#### 7.1 Sections to be defined in Dio\_MemMap.h

| Section name                               | Section type       | Description                                                         |
|--------------------------------------------|--------------------|---------------------------------------------------------------------|
| DIO_START_SEC_CODE                         | Code               | Start of Memory Section for Code                                    |
| DIO_STOP_SEC_CODE                          | Code               | End of Memory Section for Code                                      |
| DIO_START_SEC_CONFIG_DATA_↔<br>UNSPECIFIED | Configuration Data | Start of Memory Section for Config Data                             |
| DIO_STOP_SEC_CONFIG_DATA_↔<br>UNSPECIFIED  | Configuration Data | End of Memory Section for Config Data                               |
| DIO_START_SEC_VAR_INIT_32                  | Variables          | Used for initialized variables which have to be aligned to 32 bit   |
| DIO_STOP_SEC_VAR_INIT_32                   | Variables          | End of above section                                                |
| DIO_START_SEC_CONST_8                      | Variables          | Used for variables and constants which have to be aligned to 8 bit  |
| DIO_STOP_SEC_CONST_8                       | Variables          | End of above section                                                |
| DIO_START_SEC_CONST_16                     | Variables          | Used for variables and constants which have to be aligned to 16 bit |
| DIO_STOP_SEC_CONST_16                      | Variables          | End of above section                                                |
| DIO_START_SEC_CONST_32                     | Variables          | Used for variables and constants which have to be aligned to 32 bit |
| DIO_STOP_SEC_CONST_32                      | Variables          | End of above section                                                |

#### 7.2 Linker command file

Memory shall be allocated for every section defined in the driver's "<Module>\_MemMap.h.



## Chapter 8

### Integration Steps

This section gives a brief overview of the steps needed for integrating this module:

1. Generate the required module configuration(s). For more details refer to section [Files Required for Compilation](#)
2. Allocate the proper memory sections in the driver's memory map header file ("`<Module>_MemMap.h`") and linker command file. For more details refer to section [Sections to be defined in `<Module>\_MemMap.h`](#)
3. Compile & build the module with all the dependent modules. For more details refer to section [Building the Driver](#)

## Chapter 9

### External assumptions for driver

The section presents requirements that must be complied with when integrating the DIO driver into the application.

| External Assumption Req ID | External Assumption Text                                                                                                                                                                                                                                                                                            |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SWS_Dio_00061              | The Dio module shall not provide APIs for overall configuration and initialization of the port structure which is used in the Dio module. These actions are done by the PORT Driver Module. Note: DIO module implementation shall be made independent of PORT configuration. (DIO_SW001)                            |
| SWS_Dio_00102              | The Dio module's user shall only use the Dio functions after the Port Driver has been initialized. Otherwise the Dio module will exhibit undefined behavior. Note: Dio module works on pins and ports which are configured by the Port driver                                                                       |
| SWS_Dio_00127              | The Port module shall configure a DIO channel as input or output [SWS↔_Dio_00001 and SWS_Dio_00002].                                                                                                                                                                                                                |
| SWS_Dio_00001              | The Dio module shall not provide an interface for initialization of the hardware. The Port Driver performs this. Note: DIO module implementation shall be made independent of PORT configuration. (DIO_SW001)                                                                                                       |
| SWS_Dio_00002              | The PORT driver shall provide the reconfiguration of the port pin direction during runtime. Note: DIO module implementation shall be made independent of PORT configuration. (DIO_SW001)                                                                                                                            |
| SWS_Dio_00017              | For parameter values of type Dio_ChannelType, the Dio's user shall use the symbolic names provided by the configuration description. Furthermore, S↔WS_Dio_00103 applies to the type Dio_ChannelType. Note: If supported by the external application, parameter validation at production time is no more needed.    |
| SWS_Dio_00020              | For parameter values of type Dio_PortType, the user shall use the symbolic names provided by the configuration description. Furthermore, SWS↔_Dio_00103 applies to the type Dio_PortType. Note: If supported by the external application, parameter validation at production time is no more needed.                |
| SWS_Dio_00022              | For parameter values of type Dio_ChannelGroupType, the user shall use the symbolic names provided by the configuration description. Furthermore, SWS_Dio_00056 applies to the type Dio_ChannelGroupType. Note: If supported by the external application, parameter validation at production time is no more needed. |
| EA_RTD_00071               | If interrupts are locked, a centralized function pair to lock and unlock interrupts shall be used.                                                                                                                                                                                                                  |

| External Assumption Req ID | External Assumption Text                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EA_RTD_00078               | If the platform supports configuring the port pins as Input-Output then the pins for which the application intends to use Dio_FlipChannel() function at run-time should be configured as Input-Output as in this case FlipChannel() function writes the output buffer and returns the value from the input buffer. If the platform supports configuring the port pins as either Input or Output then the pins for which the application intends to use Dio_FlipChannel() function at run-time should be configured as Output and in this case FlipChannel() function writes the output buffer and returns the value from the output buffer. |
| EA_RTD_00082               | When caches are enabled and data buffers are allocated in cacheable memory regions the buffers involved in DMA transfer shall be aligned with both start and end to cache line size. Note: <b>Rationale:</b> This ensures that no other buffers/variables compete for the same cache lines.                                                                                                                                                                                                                                                                                                                                                 |
| EA_RTD_00092               | The integrator shall allocate a single EcucPartition per core or the partition in which the Dio is allocated shall be exclusively mapped to a core. Note: Internally, the Dio will use the Core ID returned by GetCoreID API to reference the appropriate global data and configuration elements, that is why a core should reference only one configured partition.                                                                                                                                                                                                                                                                        |
| EA_RTD_00093               | The application shall define EcucCoreIDs in a compact/consecutive order, starting from zero.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| EA_RTD_00096               | The application shall pass the correct initialization pointer, specific to the partition in which the driver is to be used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| EA_RTD_00106               | Standalone IP configuration and HL configuration of the same driver shall be done in the same project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| EA_RTD_00107               | The integrator shall use the IP interface only for hardware resources that were configured for standalone IP usage. Note: The integrator shall not directly use the IP interface for hardware resources that were allocated to be used in HL context.                                                                                                                                                                                                                                                                                                                                                                                       |
| EA_RTD_00108               | The integrator shall use the IP interface to build a CDD, therefore the BSWMD will not contain reference to the IP interface                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| EA_RTD_00113               | When RTD drivers are integrated with AutosarOS and User mode support is enabled, the integrator shall assure that the definition and declaration of all RTD functions needed to be called as trusted functions follow the naming convention Call<Function_Name>TRUSTED(parameter1,parameter2,...) in Integration/User code. They need to be visible in Os.h for the driver to call them. They will call RTD <Function_Name>() as trusted functions in OS specific manner.                                                                                                                                                                   |

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